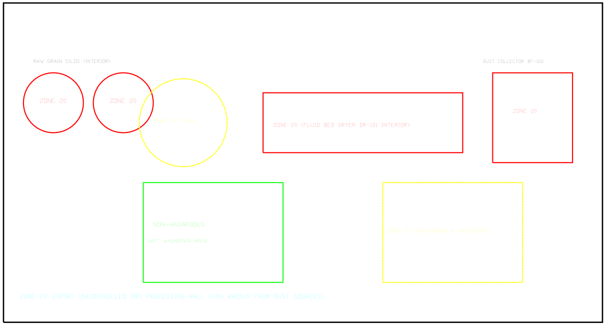
PROCESSING BUILDING - PLAN VIEW COMBUSTIBLE DUST ZONING (NFPA 652 / IEC 60079-10-2)


RAW GRAIN SILOS (INTERIOR)
ZONE 20
ZONE 20
ZONE 21 (1.5m)
ZONE 20 (FLUID BED DRYER DR-101 INTERIOR)
DUST COLLECTOR BF-102
ZONE 20
NON-HAZARDOUS
(WET WASHDOWN AREA)
ZONE 21 (PACKAGING & WEIGHING)
ZONE 22: ENTIRE UNCONTROLLED DRY PROCESSING HALL (3.0m RADIUS FROM DUST SOURCES)

DUST HAZARD DATA & COMPLIANCE:

1. Material: Toor Dal (Pigeon Pea) Grain Dust
2. Dust Deflagration Class: St 1 (Kst: 85 bar-m/s)
3. Maximum Explosion Pressure Pmax: 7.6 bar g
4. Minimum Ignition Energy (MIE): > 30 mJ
5. Minimum Ignition Temp (MIT - Cloud): 430 deg C
6. Layer Ignition Temperature (LIT - 5mm): 260 deg C
7. Permissible Equipment Surface Temp: T120 deg C
   (Max 2/3 of LIT per IEC 60079-0)
8. Zone 20 Enclosures: IP65 Ex tb IIIC T120 deg C
9. Zone 21/22 Enclosures: IP55 Ex tc IIIC T120 deg C
10. Explosion Relief: Flameless explosion vents
    on Baghouse BF-102 and Cyclone CY-101
    certified per NFPA 68 / EN 14491.
11. Static Bonding: All pipes, ductwork, filter
    cages and vessels bonded to earth (< 10 ohms).
12. Housekeeping: Scheduled dry vacuum cleaning
    preventing dust layer buildup > 1.0 mm.

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

HAZARDOUS AREA CLASSIFICATION & DUST ZONING

DISCIPLINE: SAFETY | STAGE: CONCEPTUAL FEED

DWG NO: DWG-FIN-15T-012	REV: V4.0	SHEET: 12 OF 12
SCALE: AS SHOWN DATE: 2026-10-15	STATUS: PRELIMINARY	GATE: G-A HOLD